

Lecture 4 Notes: Near Opposites: Kant and Mill

Philosophy of Mathematics

Central Theme

Kant and Mill are near opposites because both reject simple Platonism, but they explain mathematics in almost opposite ways.

Kant: mathematics is synthetic a priori.
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Mill: mathematics is synthetic a posteriori.
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The guiding contrast:

Kant	Mill
mathematics grounded in pure intuition necessary and a priori forms of possible experience	mathematics grounded in experience empirical and inductive generalizations from actual experience

1. Reorientation: Mathematics after the Scientific Revolution

By Kant's time, mathematics and science had changed dramatically.

The seventeenth century brought:

analytic geometry

calculus

Newtonian mechanics

new ideas of continuity, derivative, and limit

Mathematics was no longer only geometry and arithmetic in the ancient sense. It had become deeply connected to modern physics.

Teaching Point

The philosophy of mathematics now has to explain not only ancient geometry and arithmetic, but also the role of mathematics in modern science.

Whiteboard

Descartes Newton Leibniz

geometry + algebra + calculus + physics

New question: How does mathematics make modern science possible?

2. Rationalism and Empiricism

The background dispute is between rationalism and empiricism.

Rationalism	Empiricism
reason as source of knowledge	sense experience as source of knowledge
mathematics as model	mathematics as problem
Descartes, Spinoza, Leibniz	Locke, Berkeley, Hume, Reid

Rationalists explain the necessity of mathematics well. Mathematical truths are grasped by reason and do not depend on contingent experience.

Empiricists explain the applicability of mathematics well. Mathematical ideas arise from experience with physical objects.

But each view has a problem.

Rationalism	Empiricism
necessity explained	applicability explained
applicability mysterious	necessity mysterious

Teaching Point

Kant and Mill inherit the old Plato/Aristotle tension: exactness and necessity pull us toward rationalism, while applicability pulls us toward empiricism.

Whiteboard

Problem for rationalism: Why does pure reason apply to nature?

Problem for empiricism: Why does mathematics seem necessary?

3. Kant's Central Problem

Kant tries to synthesize rationalism and empiricism.

His problem is:

How can mathematics be a priori and yet apply universally to experience?

He wants to explain three features at once:

necessity + a priori + applicability

Kant's answer is that mathematics is *synthetic a priori* and grounded in *intuition*.

Teaching Point

Kant's philosophy of mathematics is an attempt to preserve the necessity of mathematics without making mathematics irrelevant to empirical science.

Whiteboard

Kant's task:

necessary + a priori + applicable

Mathematics = synthetic a priori

4. Analytic and Synthetic

Kant distinguishes analytic from synthetic judgements.

Analytic	Synthetic
predicate contained in subject concept known by conceptual analysis elucidatory	predicate adds something new requires intuition or experience ampliative

Example of an analytic judgement:

All bachelors are unmarried.

The predicate concept "unmarried" is already contained in the subject concept "bachelor."
Kant thinks most mathematical judgements are not like this. They are synthetic.

Teaching Point

For Kant, mathematical propositions do not merely unpack definitions. They extend our knowledge.

Whiteboard

Analytic: unpack the concept

Synthetic: add something new

Question: $7 + 5 = 12$ analytic or synthetic?

5. Kant on Arithmetic: $7 + 5 = 12$

Kant famously argues that:

$$7 + 5 = 12$$

is synthetic.

The concept of the sum of seven and five contains only the idea of combining seven and five into one number. It does not already contain the concept of twelve.

To get twelve, we must construct the sum.

$$7 \rightarrow 8 \rightarrow 9 \rightarrow 10 \rightarrow 11 \rightarrow 12$$

This construction involves intuition, especially succession in time.

Teaching Point

For Kant, arithmetic is not mere definition. It requires constructive activity in intuition.

Whiteboard

$7 + 5 \neq$ conceptual analysis alone

7, 8, 9, 10, 11, 12

Arithmetic uses succession.

6. Kant on Geometry: Construction in Intuition

Kant also thinks geometry is synthetic a priori.

A philosopher who only analyzes the concept of triangle will not discover that the angles of a triangle sum to two right angles.

The geometer does something different. She constructs a triangle and then adds auxiliary constructions.

For example, in Euclid's proof that the angles of a triangle sum to two right angles, the geometer:

constructs a triangle

extends one side

draws a parallel line

derives the angle sum

Teaching Point

Kant's model of mathematics is not passive inspection of concepts. It is active construction in pure intuition.

Whiteboard

Concept of triangle $\nrightarrow$ angle sum $= 180^\circ$

construct $\rightarrow$ extend $\rightarrow$ draw parallel $\rightarrow$ prove

7. Philosophy and Mathematics

Kant contrasts philosophical cognition with mathematical cognition.

Philosophy	Mathematics
rational cognition from concepts	rational cognition from construction of concepts
conceptual analysis	intuition and construction
makes explicit what is implicit	produces new knowledge

This is why mathematics can be synthetic and yet a priori.

It is not based on empirical observation, but it is also not based merely on definitions.

Teaching Point

Kant gives mathematics its own distinctive epistemology: neither ordinary empirical knowledge nor mere conceptual analysis.

Whiteboard

Philosophy: concepts

Mathematics: construction of concepts

Mathematics is not just logic or definition.

8. Pure Intuition: Space and Time

Kant's key idea is that mathematics is grounded in pure intuition.

Geometry $\rightarrow$ space

Arithmetic $\rightarrow$ time/succession

Space and time are not learned from ordinary experience. Rather, they are forms through which we experience objects.

Mathematics applies to experience because experience itself must be structured spatially and temporally.

Teaching Point

Kant's great move is to locate mathematics neither in a Platonic realm nor in ordinary empirical objects, but in the form of possible experience.

Whiteboard

Mathematics concerns the form of experience.

Geometry = form of space

Arithmetic = form of succession

That is why mathematics applies.

9. Kant's Strength

Kant's view explains why mathematics seems necessary, a priori, and applicable.

Feature	Kant's Explanation
necessary	experience must conform to space and time
a priori	space and time are forms of intuition, not empirical discoveries
applicable	objects of experience appear within those forms

Kant therefore tries to solve the old problem that divided rationalists and empiricists.

Teaching Point

Kant's view is powerful because it explains both the certainty of mathematics and its application to experience.

Whiteboard

Kant's success:

mathematics structures possible experience

$\therefore$ a priori and applicable

10. Kant's Problem: Non-Euclidean Geometry

Kant's view is placed under pressure by later developments in mathematics and physics.

The central example is non-Euclidean geometry.

Kant held that Euclidean geometry describes the necessary form of spatial intuition. But modern physics uses non-Euclidean geometry to describe space-time.

This creates a serious problem:

Can a supposedly a priori truth be overturned by empirical science?

If Euclidean geometry is synthetic a priori, then it should not be empirically revisable. But the development of non-Euclidean geometry and its application to physics suggests otherwise.

Teaching Point

Non-Euclidean geometry threatens Kant's claim that Euclidean geometry gives the necessary form of spatial experience.

Whiteboard

Kant: Euclidean geometry is necessary.

Modern physics: space-time may be non-Euclidean.

Problem: What happens to the synthetic a priori?

11. Further Problems for Kant

Other developments in mathematics also strain Kant's account.

Examples include:

complex analysis

higher-dimensional geometry

functional analysis

set theory

These areas do not obviously concern the forms of ordinary spatial or temporal perception.

Kant's view works best for Euclidean geometry and elementary arithmetic. It is less clear how it extends to modern mathematics.

Teaching Point

Kant gives a powerful account of traditional mathematics, but later mathematics stretches or breaks the link between mathematics and intuition.

Whiteboard

Euclidean geometry fits Kant well

set theory, complex analysis, higher dimensions fit less well

12. Mill's Radical Empiricism

Mill takes the opposite route from Kant.

He rejects a priori mathematical knowledge. Mathematics is not grounded in pure intuition. It is grounded in experience.

Mathematics is empirical.

Mill distinguishes verbal propositions from real propositions.

Verbal propositions	Real propositions
true by definition	about the world
no genuine factual content	empirical content
not mathematics	mathematics

For Mill, mathematics and much of logic consist of real propositions. They are ultimately known through induction from experience.

Teaching Point

Mill agrees with Kant that mathematics is not merely analytic. But he denies Kant's a priori intuition and makes mathematics empirical.

Whiteboard

Kant: synthetic a priori

Mill: synthetic a posteriori

Mathematics = real propositions about the world

13. Mill's Epistemology: Induction

Mill's basic model of knowledge is enumerative induction.

many observed cases $\longrightarrow$ general law

Example:

many black crows $\longrightarrow$ all crows are black

Mill applies this model to mathematics.

Mathematical propositions summarize regularities discovered in experience. We repeatedly combine and separate objects and observe stable patterns.

Teaching Point

Mill's view is naturalistic: mathematics is part of the same empirical inquiry as science.

Whiteboard

Observation $\longrightarrow$ generalization $\longrightarrow$ mathematics

No special a priori faculty.

14. Mill on Geometry

Mill rejects abstract geometrical objects.

A geometrical line is not a Platonic object or a Kantian construction in pure intuition. It is an idealization of physical lines.

drawn line $\rightarrow$ thinner $\rightarrow$ straighter $\rightarrow$ ideal line

Points, lines, and circles are limit concepts.

Strictly speaking, Euclidean geometry is fictional, because there are no perfect breadthless lines or dimensionless points in nature. But it is useful because real physical objects approximate geometrical figures.

Teaching Point

Mill explains the applicability of geometry by making geometry an idealized empirical science.

Whiteboard

Millian geometry:

idealized physical figures

perfect line = limit of thinner and straighter drawn lines

geometry is approximately true of nature

15. Mill's Problem: Exact Geometry

Mill's geometry faces a difficulty.

If geometry is based on physical experience, how can it yield exact truths?

For example:

Between any two points, a straight line can be drawn.

But physically, we cannot draw a breadthless line. Nor can we bisect a line segment indefinitely. Eventually we reach physical limits.

So Mill must appeal to idealization or possibility. But then the question becomes:

Where does this notion of possibility come from?
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If all knowledge comes from experience, how do we know what is ideally possible beyond experience?

Teaching Point

Mill explains applicability, but at the cost of making exactness difficult to understand.

Whiteboard

Experience gives approximate figures.

Geometry gives exact truths.

Problem: How does approximation yield exactness?

16. Mill on Arithmetic

Mill also treats arithmetic empirically.

Numbers are always numbers of something.

ten bodies ten sounds ten heartbeats

There are no numbers in the abstract.

Numerals are not names of abstract objects. Rather, they are general terms applying to aggregates.

“two” denotes all pairs

“twelve” denotes all dozens

Arithmetic sums are generalizations about what happens when aggregates are collected or separated.

$$2 + 1 = 3$$

means, roughly:

two things combined with one thing make three things.

Teaching Point

Mill makes arithmetic about physical aggregates. This explains why arithmetic applies to ordinary objects.

Whiteboard

Number = number of things

“two” = all pairs

$2 + 1 = 3$ from collecting and separating objects

17. Frege's Criticism: What Are the Units?

Frege objects that Mill's account cannot explain how units are determined.

A physical aggregate can be counted in different ways.

one pack of cards

four suits

fifty-two cards

The physical object alone does not determine the number. We need a concept under which we are counting.

Likewise, a bundle of straw can be counted as:

one bundle

many straws

many half-straws

many molecules

Teaching Point

Counting is not just perceiving a physical aggregate. It requires identifying units under a concept.

Whiteboard

$$\text{Same physical stuff} \rightarrow \begin{cases} 1 \text{ pack} \\ 4 \text{ suits} \\ 52 \text{ cards} \end{cases}$$

Question: What determines the units?

18. Frege's Criticism: Large Numbers

Frege also presses Mill on large numbers.

What experience confirms a sum like:

$$1,234,457,890 + 6,792 = 1,234,464,682?$$

We do not verify this by collecting billions of objects. In fact, empirical counting would be a terrible way to confirm such a sum. Human error would be overwhelming.

We know the sum by calculation, not by direct observation.

Teaching Point

Mill's inductive account works best for small arithmetic. It struggles with large numbers, infinity, and higher mathematics.

Whiteboard

Small sums: $2 + 1 = 3$

Large sums: $1,234,457,890 + 6,792 = ?$

Do we confirm this by observation?

19. Mill on Apparent Necessity

Mill does not deny that mathematics seems necessary. He explains this appearance psychologically.

Basic arithmetic and geometry are confirmed by early and constant experience. We have encountered these regularities since childhood, and we cannot imagine them failing.

apparent necessity = early and constant experience

Mathematical necessity is not genuine a priori necessity. It is the result of deeply entrenched empirical expectation.

Teaching Point

Mill replaces metaphysical necessity with psychological irresistibility formed by repeated experience.

Whiteboard

Why does math seem necessary?

Because experience has always confirmed it.

Necessity $\rightarrow$ habit + imagination

20. Mill's Limits

Mill's account has several major limits.

Problem	Why it is difficult for Mill
Exact geometry	experience gives approximations, not exact lines and points
Large arithmetic	we do not confirm large sums by observation
Mathematical induction	enumerative induction does not justify mathematical induction
Infinity	we do not experience infinite collections
Higher mathematics	set theory, analysis, and abstract mathematics exceed simple empirical construction

Teaching Point

Mill's empiricism is bold and naturalistic, but it is too weak to account for the full range of mathematics.

Whiteboard

Mill's problem:

experience is finite, approximate, and fallible

mathematics is infinite, exact, and certain?

21. Kitcher: A Modern Millian

The chapter ends by connecting Mill to Philip Kitcher.

Kitcher keeps the empiricist spirit but improves Mill's account through idealization.

Instead of actual human construction, Kitcher appeals to ideal constructors.

actual human constructors $\longrightarrow$ ideal constructors

Ideal constructors are fictional agents without ordinary human limits. They can draw breadthless lines, collect very large aggregates, and perform idealized operations.

This allows a more sophisticated empiricist account of higher mathematics.

Teaching Point

Kitcher shows how Mill's basic project can be modernized: mathematics is still connected to human constructive activity, but through idealized activity rather than actual observation.

Whiteboard

Mill: actual experience

Kitcher: idealized construction

empiricism + idealization

Conclusion: Kant and Mill as Near Opposites

Kant and Mill agree that mathematics is not merely verbal or analytic.

Both treat mathematics as substantive.

But they disagree radically about its epistemology.

Kant	Mill
synthetic a priori	synthetic a posteriori
pure intuition	enumerative induction
forms of possible experience	generalizations from experience
necessity preserved	necessity explained away
applicability through structure of experience	applicability through empirical origin

Final Framing

To teach this chapter well, present Kant and Mill as two attempts to answer the same problem:

How can mathematics be both exact and applicable?

Kant protects exactness, necessity, and a priority by grounding mathematics in pure intuition and the forms of experience. But later mathematics and physics make this view hard to sustain.

Mill protects naturalism and applicability by making mathematics empirical and inductive. But exactness, infinity, large numbers, and higher mathematics become difficult to explain.

Kant makes mathematics too a priori for modern science.

Mill makes mathematics too empirical for mathematical exactness.

Discussion Questions

1. Why does mathematics create a problem for empiricism?
2. Why does mathematics create a problem for rationalism?
3. Is $7 + 5 = 12$ analytic or synthetic?
4. What does Kant mean by saying that mathematics constructs concepts?
5. How does pure intuition explain the applicability of mathematics?
6. Does non-Euclidean geometry refute Kant's philosophy of mathematics?
7. Is Mill right that mathematics is based on experience?
8. Can arithmetic be understood as generalization from collecting and separating objects?
9. What determines the units when we count?
10. Does mathematics seem necessary only because of habit and repeated experience?
11. Is Kitcher's appeal to ideal constructors a genuine improvement on Mill?

Key Terms

Rationalism The view that reason is a fundamental source of knowledge.

Empiricism The view that sense experience is the primary source of knowledge.

Analytic judgement For Kant, a judgement in which the predicate concept is contained in the subject concept.

Synthetic judgement For Kant, a judgement in which the predicate concept adds something not already contained in the subject concept.

A priori Knowable independently of particular experience.

A posteriori Knowable through experience.

Synthetic a priori A judgement that extends knowledge but is knowable independently of particular experience.

Intuition For Kant, a singular representation of an object; in mathematics, tied to construction in space and time.

Pure intuition A priori intuition of the forms of possible experience, especially space and time.

Construction of concepts Kant's account of mathematical cognition, in which a concept is exhibited in intuition.

Transcendentalism The view that philosophy investigates the conditions or limits of possible experience prior to empirical science.

Naturalism The view that philosophy is continuous with science and does not stand above it as first philosophy.

Verbal proposition Mill's term for a proposition true by definition and lacking genuine factual content.

Real proposition Mill's term for a proposition with factual content about the world.

Enumerative induction Inference from observed cases to a generalization.

Limit concept A concept approached by idealizing a sequence of empirical approximations, such as thinner and straighter lines.

Ideal constructor Kitcher's fictional idealized agent capable of mathematical constructions beyond actual human limitations.